What Can Missing-Modality Evaluations Identify under Selective Observation? A Finite-Support Analysis

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Abstract

Models that consume several input groups ("modalities") are usually compared by deleting groups at random from *complete* test cases. We ask what such an evaluation, together with what an evaluator can actually observe, identifies about the risk difference of two fixed predictors under the natural, possibly informative, missingness of deployment. For finite supports we write each observation model as linear constraints on the joint law of cells and masks, and combine it with a pattern-mixture sensitivity model relative to the complete-case law. Three findings follow. First, with labels only on complete cases and a sensitivity parameter above the true one (so away from the MCAR boundary), the natural-missingness Brier difference is point-identified from complete cases plus the unlabelled law of incomplete units only if the two predictors agree on every incomplete input that occurs. Without mask frequencies the sign is not strictly identified even under MCAR: with complete cases alone A is only weakly better, and if the strata's conditional laws are also given, strictly better for every valid law but with no uniform margin. Second, random dropout on complete cases can rank the predictors opposite to the natural-missingness truth. Third, we ran a semi-synthetic repair-validation pilot: two grouped tabular features of a public census dataset, a covariate-profile split without train/evaluation overlap, and a fixed synthetic label-dependent mask rule integrated exactly. Complete-case dropout again ranks the predictors opposite to the risk difference under the mask rule ($+0.0004$ vs. -0.0028 Brier). The exact-observable identified sets admit both orders at every non-empty value of the sensitivity parameter. The effects are small, the masks are synthetic, and no real modality missingness is analysed.

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1. Introduction

Robustness to missing modalities is typically measured by *artificial* deletion: an input group is dropped at random from complete test cases and the risks of competing models are compared (Neverova et al., 2016; Ma et al., 2022). At deployment, groups are missing for reasons that can depend on the missing values and on the outcome. The usual evaluation conflates two masks, the artificial M drawn by the evaluator independently of the data and the natural R whose law is unknown. It also conflates two populations: the complete cases on which M is applied, and the full population in which R occurs. Under informative missingness $P(C \mid R = \mathbf{1}) \neq P(C)$.

We study one question: *given what an evaluator can observe, what does a missing-modality evaluation identify about the natural-missingness risk difference of two fixed predictors?* We propose neither an estimator nor an optimisation method. Finite-support partial identification by linear programming is classical (Manski, 2003; Balke & Pearl, 1997; Horowitz & Manski, 2000). Bounded likelihood-ratio sensitivity models are standard (Rosenbaum, 1987; Tan, 2006; Zhao et al., 2019; Dorn & Guo, 2023). Bounding a performance difference directly, rather than two separate risks, is known to cancel shared uncertainty (Guerdan et al., 2024). The closest evaluation work (Subbaswamy et al., 2021) bounds each model's worst-case risk under a label-dependent measurement-policy shift, but not from the information available when features are partly missing.

Our contribution is an account of what each observation model identifies in this problem:

- **Observation models and identification.** We separate four information sets and derive each observable as a linear row. We give the exact linearisation of the sensitivity model and the positivity conditions under which a solution is a valid law rather than a limit. We characterise when the difference is identified: with complete cases and unlabelled incomplete units, and

a sensitivity parameter above the true one, the Brier difference is identified only when the predictors agree on the incomplete inputs that occur. We also show how the conclusion at the MCAR boundary depends on which strata are known to exist (Sections 3 and 4).

- **Dropout versus truth.** Complete-case random dropout can reverse the natural-missingness ranking, and a sensitivity parameter below the truth can yield a non-empty set that certifies the wrong sign (Section 5).

- **Where the uncertainty comes from.** On a fixed semi-synthetic pilot we compare the realised sample with the noise-free expected law. This separates ambiguity caused by missing information from ambiguity caused by mask and sampling noise (Section 7.1).

The toy results are exact but come from one hand-specified population. The pilot uses two grouped tabular features (education years, weekly hours) treated as "modalities" and synthetic masks. Neither says how often rankings change in practice.

2. Related Work

Missingness shift. Zhou et al. (2023) study domain adaptation when source and target share $P(X, Y)$ but differ in the missingness mechanism, with identification results for MCAR-type shifts. The preprint of Rockenschaub et al. (2024) shows that the Bayes predictor is invariant only under ignorable shifts and that label-dependent shifts are not ignorable. Choi et al. (2026) show that estimates of a modality's value are biased when modality availability differs between source and target and correct them by inverse probability weighting under a missing-at-random assumption. These works learn or re-weight under ignorability; we study non-ignorable (value- and label-dependent) missingness through partial identification and do not learn predictors.

Evaluation under shift. Subbaswamy et al. (2021) define the worst-case risk of a fixed model when a user-chosen conditional $P(W \mid Z)$ shifts, as the expected loss on the worst $(1-\alpha)$ subsample, and estimate it by a conditional-quantile (CVaR-type) dual. Their sepsis case study shifts $P(\text{test order} \mid \text{demographics, disease status})$, i.e. a label-dependent measurement policy, and reports the worst-case accuracy of two models separately (a baseline score is also evaluated on one model's worst subsamples, a same-subsample comparison, but no difference is bounded under a common shift). Our setting can be written in their notation with $W = R$ and $Z = (X, Y)$; we differ by (i) conditioning the information on what is observable when X is partly missing, (ii) using a two-sided pattern-mixture ratio instead of a one-sided subsample bound, and (iii) bounding the difference of two models. Guerdan et al. (2024) bound the performance difference (a "δ-regret interval") between a status-quo decision policy and a proposed one under selective labels, using marginal-sensitivity-type bounding functions, and show that terms shared by both policies cancel; their minimal uncertainty set is stated given the bounding functions and they note that the resulting interval need not be the tightest implied by the causal assumption. The joint-versus-separate comparison in our experiments is an instance of their observation and is not claimed as new. Section E tabulates, for these two papers and ours, what is observed, what is compared, the sensitivity model, the sharpness claim and the treatment of sampling uncertainty.

Partial identification and sensitivity analysis. Sharp bounds with missing outcomes and covariates go back to Manski (Manski, 2003; Horowitz & Manski, 2000); linear-programming bounds for discrete variables are standard (Balke & Pearl, 1997). Our sensitivity model is a pattern-mixture restriction (Little, 1993) of the bounded-likelihood-ratio type used by Rosenbaum (1987); Tan (2006); Zhao et al. (2019); Dorn & Guo (2023) and, for biased samples, by Sahoo et al. (2022). The ranking-transfer threshold we report is a tipping-point summary in the spirit of the E-value (VanderWeele & Ding, 2017). Confidence sets for partially identified parameters are discussed by Imbens & Manski (2004); Stoye (2009); our outer interval is a simple, conservative projection construction.

Adversarial deletion and missing labels. Worst-case per-instance feature deletion is studied for learning by Globerson & Roweis (2006). Non-random missingness of ratings changes the apparent ranking of recommender models (Marlin & Zemel, 2009), and robust policy improvement under confounding is studied by Kallus & Zhou (2018). These concern learning or missing labels rather than evaluation with missing features.

3. Problem Setup: Observation Models

Population and masks. A unit has a cell $c = (x, y) \in \mathcal{C}$ with $x \in \{0,1\}^d$ and $y \in \{0,1\}$, a natural mask $R \in \mathcal{R} = \{0,1\}^d$ ($r_j = 1$ means modality j is observed, $\mathbf{1}$ is the complete mask), and a label indicator $S \in \{0,1\}$. The unknown joint mass is

$$p(c, r) = P(C = c, R = r),$$
$$w_c = \textstyle\sum_r p(c, r), \qquad \rho_r = \textstyle\sum_c p(c, r),$$

where w is the full-data law and ρ the mask law. The evaluator's artificial mask $M \sim q$ is drawn independently of (C, R). Two fixed predictors f_A, f_B see x_r (the observed coordinates) and r; $L_f(c, r) = \ell(f(x_r, r), y)$ and $D(c, r) = L_A(c, r) - L_B(c, r)$. We use the Brier loss, so $D(c, r) = u(x_r) - 2y\,\delta(x_r)$ with $\delta = f_A - f_B$ and $u = f_A^2 - f_B^2$.

Estimands. The target is the natural-missingness risk difference

$$\Delta_{\text{nat}} = \sum_{c,r} p(c,r)\, D(c,r).$$

The usual evaluation computes the random-dropout difference on complete cases, $\Delta_{\text{cc}}^{\text{drop}}(q) = \sum_c v_c \sum_r q_r D(c,r)$ with $v_c = P(C = c \mid R = \mathbf{1})$. We write $\Delta_r^{\text{cc}} = \sum_c v_c D(c,r)$ for the per-mask differences on complete cases.

Assumption 1 (Labels)**.** $S = \mathbb{1}[R = \mathbf{1}]$: labels are observed exactly for complete cases. (If label availability were independent of the mask, S would enter as a separate factor; we do not analyse that case.)

Information sets. Each observable becomes one family of linear rows in the unknowns. Constants are the observed quantities; unknowns are p and the auxiliary scalars listed.

- **B** complete cases only (known: v; unknown: p, $\rho_{\mathbf{1}}$): $p(c,\mathbf{1}) - \rho_{\mathbf{1}} v_c = 0$ for all c; $\sum_{c,r} p(c,r) = 1$.
- **C** one sample with Assumption 1 (known: $\theta_c = p(c,\mathbf{1})$ and $\omega(r,o) = P(R = r, X_r = o)$ for $r \neq \mathbf{1}$): $p(c,\mathbf{1}) = \theta_c$; $\sum_{c:\, x_r(c)=o} p(c,r) = \omega(r,o)$. These rows fix all ρ_r.
- **D** separate samples (known: v and $\mu_r(o) = P(X_r = o \mid R = r)$; unknown: p, ρ): $p(c,\mathbf{1}) - \rho_{\mathbf{1}} v_c = 0$; $\sum_{c:\, x_r(c)=o} p(c,r) - \mu_r(o)\rho_r = 0$.
- **A** oracle: the rows of C plus $\sum_r p(c,r) = w_c$ with w known. A is not available in practice; it is the observation model implicitly assumed when complete data are treated as the population.

In the notation above, B $\supseteq$ D $\supseteq$ C $\supseteq$ A as feasible sets. Replacing $P(C \mid R = \mathbf{1})$ by $P(C)$ moves the analysis from C to A without justification; our experiments show the change is material.

Sensitivity model. For each incomplete mask r, the full-data law among units with mask r may differ from the complete-case law by a bounded likelihood ratio:

$$\frac{\lambda_r}{\Gamma} \leq \frac{P(C = c \mid R = r)}{v_c} \leq \lambda_r \Gamma \quad \text{for all } c \text{ with } v_c > 0,$$
$$P(C = c \mid R = r) = 0 \quad \text{if } v_c = 0, \tag{1}$$

for some unknown $\lambda_r > 0$; equivalently $P(\cdot \mid R = r)$ is absolutely continuous with respect to the complete-case law and the ratio varies across cells by at most Γ^2. (v1 stated the ratio bound only for $v_c > 0$ and omitted the absolute-continuity clause, which the linear program always imposed.) We write Γ_{cc} for this parameter. It is *not* the parameter Γ_{box} of the earlier oracle analysis (Section C), whose reference is the dropout law q (equivalently the full-data law w) with a fixed scale; the two numbers must not be compared. We call this $\mathcal{M}_{\text{cc}}(\Gamma)$. $\Gamma = 1$ means every mask has the complete-case law, i.e. the mask is independent of the cell (MCAR). Γ is chosen by the analyst and is not estimable.

Lemma 1 (Exact linearisation)**.** *Multiplying (1) by ρ_r gives $\rho_r \lambda_r v_c/\Gamma \leq p(c,r) \leq \Gamma \rho_r \lambda_r v_c$, which contains the product of two unknowns. Because λ_r appears in no other constraint, the map $(\rho_r, \lambda_r) \mapsto b_r = \rho_r \lambda_r$ is onto $[0,\infty)$ whenever $\rho_r > 0$, and $p(\cdot, r) = 0$ when $\rho_r = 0$. Hence*

$$b_r \kappa_c/\Gamma \leq p(c,r) \leq \Gamma b_r \kappa_c, \qquad b_r \geq 0 \tag{2}$$

with the known constant $\kappa = v$ (B, D) or $\kappa = \theta$ (A, C) describes the same set of p among laws with $\rho_{\mathbf{1}} > 0$. No unknown product is plugged in. Reverse construction: from a feasible (p, b) with $\rho_{\mathbf{1}} > 0$, set $v = p(\cdot,\mathbf{1})/\rho_{\mathbf{1}}$ and, for masks with $\rho_r > 0$, $\lambda_r = b_r/\rho_r$ (times $\rho_{\mathbf{1}}$ when $\kappa = \theta$); then (1) holds, including absolute continuity. If $\rho_{\mathbf{1}} = 0$ the point is not a probability model of the data (no complete cases) and is only a limit of such models.

The identified set of p is therefore the projection of a polytope, and Δ_{nat} ranges over an interval computed by two linear programs. Section A records one caveat: in B and D the programs admit $\rho_{\mathbf{1}} = 0$, which is not a valid population (no complete cases), so the computed interval is the *closure* of the identified set.

Two relaxations of (2) must be kept apart. SUPPORT-ONLY keeps only $p(c,r) = 0$ whenever $\kappa_c = 0$; it is the closure of $\bigcup_{\Gamma<\infty} \mathcal{M}_{\text{cc}}(\Gamma)$. NO-MODEL drops all restrictions (Manski-type). They differ exactly when the complete-case support misses a cell.

Reading an interval. For $\Delta = R_A - R_B$ with identified interval $[\ell, u]$ we distinguish: *strict* ($u < 0$ or $\ell > 0$; margin $\min(|\ell|, |u|)$); *weak* ($u = 0$ or $\ell = 0$ and the zero endpoint is attained by a valid model: one model is never worse and a tie is compatible with the data); *no-margin* (the zero endpoint is reached only in the closure: strict for every valid model but without a uniform margin); *both orders* ($\ell < 0 < u$; interior values are always attained by valid models because the valid set is convex); and *tie*. Write P_Γ for the closed relaxed polytope, R_+ for the strata whose positive mass the setting's information presupposes ($\rho_{\mathbf{1}}$ in B; $\rho_{\mathbf{1}}$ and every given stratum in D; none in A and C), and $F_\Gamma = \{x \in P_\Gamma : \rho_r(x) > 0 \ \forall r \in R_+\}$ for the valid laws. The order of the analysis is fixed: (1) a linear program maximises $\min_{r \in R_+} \rho_r$ over P_Γ; if the value is 0 the valid set is empty, no interval is reported, and we write $\emptyset^c$ (the relaxed program is feasible but no point of it is a valid law); (2) only otherwise are the extrema of Δ computed, and by the standard convexity argument ($x_\epsilon = (1-\epsilon)x + \epsilon x_+$) they are the infimum and supremum over F_Γ; (3) each endpoint

is then classified as attained or closure-only by the same maximisation with the objective fixed at the endpoint. All three steps are exact rational programs with certificates; only the Hoeffding boxes of Section 6 carry floating-point inputs.

Two thresholds. For a hidden joint p^* let $\Gamma_{\mathrm{true}} = \inf\{\Gamma : p^* \in \mathcal{M}_{\mathrm{cc}}(\Gamma)\}$. For an observable law o let $\Gamma_{\min}(o) = \inf\{\Gamma : \mathcal{M}_{\mathrm{cc}}(\Gamma)$ contains a joint consistent with $o\}$. If p^* generated o then $\Gamma_{\min}(o) \le \Gamma_{\mathrm{true}}$; the gap consists of other hidden joints that explain the same observables. Feasibility at some $\Gamma \ge \Gamma_{\min}(o)$ says nothing about Γ_{true}, and infeasibility at Γ is an exact refutation of $\mathcal{M}_{\mathrm{cc}}(\Gamma)$ for the law o, which is not a level-α rejection (Section 6).

4. Identification

Let F be the lifted feasible polytope in (p, b, ρ) and g the objective, with zeros on auxiliary coordinates.

Theorem 2 (General criterion). *Δ_{nat} is constant on F if and only if $g \perp \mathrm{span}\{z - z' : z, z' \in F\}$. Equivalently, g lies in the row space of the equality constraints augmented with $e_j^\top$ for every coordinate j that is constant on F.*

This is a standard polyhedral fact (the affine hull of F is cut out by its equalities and implicit equalities); we state it because the familiar row-space test *without* the implicit equalities is only sufficient. It becomes necessary under a relative-interior assumption: some $z \in F$ lies strictly inside every non-degenerate bound. That assumption fails in three cases that matter here. (a) At $\Gamma = 1$ every incomplete coordinate is tied to b_r; in our population the explicit test reports "not identified" while the width is exactly zero (Table 3). (b) With structural zeros, where $\kappa_c = 0$ forces $p(c, r) = 0$. (c) At the smallest Γ for which a set is feasible, bounds may bind at every feasible point.

Proposition 3 (MCAR boundary). *Under $\mathcal{M}_{\mathrm{cc}}(1)$, $P(C \mid R = r) = v$ for every r with $\rho_r > 0$, so $\Delta_{\mathrm{nat}} = \sum_r \rho_r \Delta_r^{\mathrm{cc}}$. (i) In C, Δ_{nat} is point-identified, and $\mathcal{M}_{\mathrm{cc}}(1)$ is refutable: it is feasible iff $\omega(r, o) = \rho_r \sum_{c : x_r(c) = o} v_c$ for all (r, o). (ii) The valid laws differ between B and D. In B only the complete stratum must exist, so the identified set is $\{\sum_r \rho_r \Delta_r^{\mathrm{cc}} : \rho \in \Delta(\mathcal{R}), \rho_{\mathbf{1}} > 0\}$, and $\mathcal{M}_{\mathrm{cc}}(1)$ is never refuted. In D the conditional law μ_r of every incomplete stratum is part of the information, which presupposes $\rho_r > 0$ for each such stratum. Under $\mathcal{M}_{\mathrm{cc}}(1)$ the D rows force $b_r\left(\sum_{c \in G(r,o)} v_c - \mu_r(o)\right) = 0$, so a valid law exists iff every given μ_r equals the complete-case marginal of x_r; otherwise D refutes $\mathcal{M}_{\mathrm{cc}}(1)$ although the LP closure is non-empty. When valid laws exist, the set is $\{\sum_r \rho_r \Delta_r^{\mathrm{cc}} : \rho_r > 0$ for all given strata$\}$. Both sets have closure $[\min_r \Delta_r^{\mathrm{cc}}, \max_r \Delta_r^{\mathrm{cc}}]$, but different endpoints are attained. If $\Delta_{\mathbf{1}}^{\mathrm{cc}} = 0$ and $\Delta_r^{\mathrm{cc}} < 0$ for every incomplete r, then in B A is* weakly *better (the tie is attained at $\rho_{\mathbf{1}} = 1$), while in D, when valid laws exist, A is strictly better for every valid law but has no uniform margin (the tie is a closure point). The mirror statement holds with the signs reversed. (iii) $\Delta_{\mathrm{cc}}^{\mathrm{drop}}(q) = \Delta_{\mathrm{nat}}$ at $\Gamma = 1$ iff $\sum_r (\rho_r - q_r)\Delta_r^{\mathrm{cc}} = 0$.*

Remark 4 (Feasibility in B and D; the observable threshold). The programs of B and D are feasible for every Γ (take $\rho = e_{\mathbf{1}}$, $p(\cdot, \mathbf{1}) = v$, no incomplete mass), so LP feasibility there carries no information; only the valid-set check does. For settings C and D the smallest sensitivity parameter compatible with the observables has a closed form. Here $\mu_r(o) = P(X_r = o \mid R = r)$ and $V_r(o) = \sum_{c \in G(r,o)} v_c$ are conditional quantities, the maximum runs over incomplete strata with $\rho_r > 0$ (unsampled strata impose nothing), the scale $\lambda_r > 0$ is free, $\Gamma \ge 1$, and $\sum_c P_r(c) = 1$. With $r_r(o) = \mu_r(o)/V_r(o)$ over the groups with $V_r(o) > 0$ and $\mu_r(o) > 0$,

$$\Gamma_{\min}(o)^2 = \max_{r : \rho_r > 0} \frac{\max_o r_r(o)}{\min_o r_r(o)}.$$

If some group has $V_r(o) = 0 < \mu_r(o)$ the supports are incompatible and no finite Γ fits; if $V_r(o) > 0 = \mu_r(o)$ inside a sampled stratum, the lower density bound $P_r(c) \ge \lambda_r v_c/\Gamma > 0$ makes the group mass positive, so again no finite Γ fits; groups with $V_r(o) = \mu_r(o) = 0$ are outside the relevant support and form no ratio. In the remaining case the infimum is attained and $\Gamma_{\min}(o) \le \Gamma_{\mathrm{true}}$ (Section A).

Proposition 5 (Setting C: identification requires agreement on incomplete inputs). *In setting C, assume $\theta_c > 0$ for all c, $\rho_r > 0$ for every incomplete mask r, and Γ^2 strictly larger than the true cross-cell spread $\max_r \left(\max_c e_r(c)/\min_c e_r(c)\right)$ with $e_r(c) = p^*(c, r)/\theta_c$; then p^* lies strictly inside the box (2) for suitable scales b_r^*. Then Δ_{nat} is identified iff, for every incomplete mask r and observed value o, $D(\cdot, r)$ is constant on the group $G(r, o) = \{c : x_r(c) = o\}$. Under the Brier loss every group contains both labels (because $\theta_c > 0$), and the condition holds iff $f_A(o, r) = f_B(o, r)$ for every incomplete input (o, r). In that case $\Delta_{\mathrm{nat}} = \sum_c \theta_c D(c, \mathbf{1})$ is a complete-case quantity. The spread hypothesis says $\Gamma > \Gamma_{\mathrm{true}}$; it concerns the hidden law and cannot be checked from the observables. Masks with $\rho_r = 0$ carry no mass and impose no condition. For $\Gamma \in [\Gamma_{\min}(o), \Gamma_{\mathrm{true}}]$, at $\Gamma = 1$, or with empirical zeros, the interiority fails: zero width then does not imply agreement, and only Theorem 2 applies.*

Remark 6 (Oracle A). Adding $\sum_r p(c, r) = w_c$ enlarges the row space to functions $a(c) + b(r, x_r(c))$ on incomplete coordinates. Under the Brier loss, with both labels at every x and the all-missing mask present, identification then holds iff δ is constant on incomplete inputs. The oracle is strictly more informative than C. In the FR1 analysis, which assumed w known and no labels at all, the analogous condition was that δ is constant on *all* inputs (Section C).

Proposition 7 (Support restriction versus $\Gamma \to \infty$). *If $p^*(c, r) > 0$ for some incomplete r and a cell with $\kappa_c = 0$,*

then $p^* \notin \mathcal{M}_{\mathrm{cc}}(\Gamma)$ *for every* Γ *and also* $p^* \notin$ SUPPORT-ONLY. *Only* NO-MODEL *contains* p^*. *Increasing* Γ *never repairs a positivity violation.*

Proofs are in Section A. They are elementary and have been checked only by the authors; the exact certificates attached to each linear program verify individual instances and are not proofs of these statements.

5. Counterexamples and Failure Modes

Example 8 (Identical unlabelled data, opposite rankings; w known)**.** Take $d = 1$, $P(X = 1) = 1/2$, $P(Y = 1 \mid X = x) = 1/4$ or $3/4$, and random dropout rate $1/4$. Let f_A be Bayes under MCAR ($f_A(\varnothing) = 1/2$) and f_B impute the mode ($f_B(\varnothing) = 1/4$). Then $D = 0$ on complete inputs, $D((x, 0), \varnothing) = 3/16$ and $D((x, 1), \varnothing) = -5/16$. Among policies that reproduce the MCAR unlabelled law, the risk difference ranges over exactly $[-5/64, 3/64]$, and random dropout gives $-1/64$: both rankings occur with identical unlabelled data.

Complete-case dropout can rank the models the wrong way. In our population (Section 7), environment E2m makes modality 2 more likely to be missing when $Y = 0$, with the same full-data law and mask frequencies as MCAR. Its natural-missingness difference is $\Delta_{\mathrm{nat}} = +0.0075$ (model B better), while random dropout on complete cases gives -0.0436 (A better) and random dropout on the population gives -0.0253 (Table 7). The reversal is one hand-specified case, not a frequency.

A too-small Γ can certify the wrong sign. For E2m the true parameter is $\Gamma_{cc} \approx 2.32$. In setting C with $\Gamma = 3/2$ the feasible set is non-empty and its interval $[-0.0495, -0.0239]$ certifies "A better". With the oracle information and $\Gamma = 2$ the interval $[0.0018, 0.0025]$ has the correct sign but excludes the truth. Feasibility therefore does not validate Γ; non-emptiness is only a necessary condition (Section B).

Support. In a constructed check, one cell is never fully observed. Under every $\mathcal{M}_{\mathrm{cc}}(\Gamma)$, including $\Gamma = 1000$, and under SUPPORT-ONLY, the oracle setting becomes infeasible. In setting C these sets stay feasible and contain the true value although the true law is excluded. NO-MODEL contains it in both settings (Table 4).

6. Sampling Uncertainty

The intervals above are population identified sets. With a finite sample, the plug-in interval (sample frequencies treated as exact) is an estimate, not a confidence set. In the FR1 analysis the plug-in interval contained the population interval in only 8/100 ($n = 100$) and 5/100 ($n = 1000$) replications. Those numbers are containment frequencies of an estimate, not coverage of a confidence procedure.

Common confidence region. All settings are functions of one observed-data law P_{obs} (here the multinomial of observable cells). For each Γ let $\mathcal{I}(\Gamma; P)$ be the identified set of p given $P_{\mathrm{obs}} = P$, and let $\mathcal{O}(\Gamma; \mathcal{R})$ be a set with $\mathcal{O}(\Gamma; \mathcal{R}) \supseteq \bigcup_{P \in \mathcal{R}} \mathcal{I}(\Gamma; P)$ for a confidence region $\mathcal{R}$. On the single event $\{P_{\mathrm{obs}} \in \mathcal{R}\}$, $\mathcal{I}(\Gamma; P_{\mathrm{obs}}) \subseteq \mathcal{O}(\Gamma; \mathcal{R})$ for every Γ at once; this is the whole coverage argument. It concerns identified sets: it does *not* validate the sensitivity assumption. If $\mathcal{M}_{\mathrm{cc}}(\Gamma)$ is false, $\mathcal{I}$ may omit the truth or be empty, and the interval can miss Δ_{nat} with any probability. The only data-based statement about Γ is one-sided: if $\mathcal{O}(\Gamma; \mathcal{R}) = \emptyset$, the hypothesis $p^* \in \mathcal{M}_{\mathrm{cc}}(\Gamma)$ (i.e. $\Gamma \geq \Gamma_{\mathrm{true}}$) is rejected at level α (under that hypothesis and $P_{\mathrm{obs}} \in \mathcal{R}$ the true law is feasible). Non-emptiness is not evidence for Γ. This level-α rejection is distinct from the exact infeasibility of $\mathcal{M}_{\mathrm{cc}}(\Gamma)$ for the known law of a finite population (exact: $\Gamma < \Gamma_{\min}(o)$, Section 7.1), which is a statement about that population only, and from the plug-in infeasibility of a single sample, which is neither.

Proposition 9 (Outer confidence interval)**.** *In setting C, assume* n *i.i.d. units with Assumption 1, so that the observable law is a multinomial over the* K *cells* $\{(c, \mathbf{1})\}$ *and* $\{(r, o) : r \text{ incomplete}\}$, K *counting every cell including those with zero count, and* $\sum_c \theta_c > 0$. *Let* B_α *be the box* $|\hat{\vartheta}_k - \vartheta_k| \leq \varepsilon$ *with* $\varepsilon = \sqrt{\log(2K/\alpha)/(2n)}$, *so that each of the* K *Hoeffding events has probability at most* $2\exp(-2n\varepsilon^2) = \alpha/K$, *rounded outward to a* 10^{-6} *grid (the implementation first adds* 10^{-9} *to* ε, *which only enlarges the box). Relax (2) to* $b_r \kappa_c^{\mathrm{lo}}/\Gamma \leq p(c, r) \leq \Gamma b_r \kappa_c^{\mathrm{hi}}$ *and bound* $b_r \leq \Gamma / \sum_c \kappa_c^{\mathrm{lo}}$. *On the event that* B_α *contains the true probabilities, which has probability at least* $1 - \alpha$, *the relaxed program contains every point of the population identified set. Its interval therefore covers the identified interval* $\mathcal{I}(\Gamma; P_{\mathrm{obs}})$ *with probability at least* $1 - \alpha$, *simultaneously for all* Γ. *On the same event it covers* Δ_{nat} *for every* $\Gamma \geq \Gamma_{\mathrm{true}}$, *i.e. whenever the true joint satisfies* $\mathcal{M}_{\mathrm{cc}}(\Gamma)$*; for* $\Gamma < \Gamma_{\mathrm{true}}$ *nothing is claimed.*

Zeros and stratum sizes. The box is built on *joint* cell probabilities with the fixed total n, so the random numbers of complete and incomplete units (random stratum sizes) need no conditioning. An empirical zero ($\hat{\theta}_c = 0$) gets $\kappa_c^{\mathrm{lo}} = 0 < \kappa_c^{\mathrm{hi}}$ and is *not* turned into a structural zero; if all $\kappa_c^{\mathrm{lo}} = 0$ (e.g. $n_{\mathrm{complete}} = 0$) the relaxed set keeps only $\{p(c, r) = 0 : \kappa_c^{\mathrm{hi}} = 0\}$, and since the Hoeffding box has every $\kappa_c^{\mathrm{hi}} \geq \varepsilon > 0$, the outer program then coincides with NO-MODEL on the incomplete masks (v1 code raised an error here). The *plug-in* program, by contrast, treats $\hat{\theta}_c = 0$ as structural and is infeasible by construction when $n_{\mathrm{complete}} = 0$; that outcome is a property of the implementation, not evidence that the data refute MCAR, and no estimand is defined there. The plug-in interval is an estimate, never a confidence statement.

The construction is conservative. It is not sharp, and we have not analysed its width or compared it with the procedures of Imbens & Manski (2004); Stoye (2009). Validity needs i.i.d. sampling, Assumption 1, and a correctly specified $\mathcal{M}_{\mathrm{cc}}(\Gamma)$.

7. Experiments: One Finite Population

Design. Everything is exact (rational arithmetic, a bounded-variable simplex with primal–dual and Farkas certificates). There is one population: $d = 2$; $P(x) = (2/5, 3/20, 3/20, 3/10)$ for $x \in \{00, 01, 10, 11\}$; $P(Y = 1 \mid x) = (1/10, 3/5, 2/5, 9/10)$. Model A is the Bayes predictor under MCAR dropout, an idealised dropout-trained model. Model B imputes the mode and applies the full-data Bayes predictor. The dropout law q drops modalities independently with rates $1/5$ and $3/10$. The environments are $\pi(r \mid c) = q_r(1 + \varepsilon(s(c) - \bar{s})(t(r) - \bar{t}))$, where s is the value of modality 2 (E1, E3) or the label (E2) and t indicates that modality 2 is missing. They share $P(C)$ and all mask frequencies with MCAR. E4 is correlated MCAR. The population was fixed before any analysis, but results from it were seen during development (a smoke test preceded the first pre-registered run), so all experiments are **exploratory**. No other population was searched. Runs: P2-FR1 (9.9 s) and P2-FR2 (33.3 s), both on 2 CPU threads.

What each information set identifies (Table 1).

- Setting C (the realistic one-sample case) point-identifies Δ_{nat} at $\Gamma = 1$. The MCAR sign is kept at $\Gamma = 1.4463$ and lost by $\Gamma = 1.4473$ (the bisection bracket), and without a sensitivity model the interval is $[-0.111, 0.078]$.
- Settings B and D give the same numbers, $[-0.130, 0]$, at $\Gamma_{cc} \in \{1, 5/4\}$, but different conclusions. In B, A is weakly better (the tie is attained at $\rho_{\mathbf{1}} = 1$). In D, the tie needs the given incomplete strata to have zero mass, so A is strictly better for every valid law but without a uniform margin. From $\Gamma_{cc} = 3/2$ both orders are compatible in B and D.
- The oracle A certifies "A better" even with no model, at $[-0.051, -0.002]$. Assuming the full-data law is known therefore changes the conclusion, which is why the FR1 results (Section C) should be read as oracle results.

The direct-difference interval is 1.6–3.5 times narrower than the separate-risk interval. This is consistent with Guerdan et al. (2024) and is not a new finding.

Dropout estimands versus the truth (Table 7, App. D). Random dropout on the population gives -0.0253 in every environment that preserves $P(C)$, because it does not depend on the policy. Random dropout on complete cases does depend on it, but it can move in the wrong direction: E2p has truth -0.058 and complete-case dropout -0.007, while E2m has truth $+0.0075$ and complete-case dropout -0.044.

Sampling (Table 8, App. D). Under $\Gamma = 1$ the plug-in program is infeasible in all six draws, although the population satisfies $\Gamma = 1$: sampling noise alone makes the observed law inconsistent with the complete-case law. A plug-in feasibility check would therefore wrongly "refute" MCAR. The outer 95% interval preserves the population sign at $\Gamma = 1$ only for $n = 10^5$ (both draws) and is undetermined at $n \le 10^4$. At $\Gamma = 2$ the population interval already contains 0. Coverage is argued by Proposition 9; we did not estimate it by Monte Carlo.

Earlier oracle results (P2-FR1). Under the FR1 sensitivity model (box around q with parameter Γ_{box}, w known), the per-mask stress tests all favour A, yet the ranking is guaranteed only up to $\Gamma^* \approx 1.44$ with pattern rates matched. Matching pattern rates barely helps over no constraint (1.40), the unlabelled law helps more (2.07), and a small hand-specified MNAR grid at $\Gamma = 2$ spans $[-0.058, -0.025]$ while the sharp interval is $[-0.071, 0.019]$. Details and all FR1 checks are in Section C. FR1 and FR2 use different sensitivity models, so their Γ values are not comparable.

7.1. Semi-synthetic pilot with grouped tabular features

Data, units and split (P2-PILOT-CLEAN1). We use UCI Adult (Becker & Kohavi, 1996) (CC BY 4.0). Of 48,842 parsed records, 44,355 were selected in an earlier development run, each with weight 1. The population is therefore an unweighted record population, not the survey-weighted census population. The earlier development split hashed a key that contained the raw income string, i.e. the target; as a result, 956 covariate profiles fell on both sides (Section G). The repair keeps the same 44,355 records. It parses labels strictly: the trailing "." is removed only from the allowed tokens, and no token was unresolved. It then hashes the canonical non-label covariate profile, i.e. all attributes except the survey weight and income, with integers in canonical form. The hash rule and the 50/50 ratio are unchanged. This yields 41,439 profile groups, none in both splits (22,188 train and 22,167 evaluation records). 2,475 groups have several records, and 1,029 of them have conflicting labels; these need not be errors, since distinct records can share an observed covariate profile. This is a record-profile split, not person isolation. The repair was run once on already-seen data; it is a repair validation, not an independent confirmation.

Design (unchanged rules). The two "modalities" are two grouped tabular features, coarsened by fixed thresholds to three levels: education years (≤ 9, 10–12, ≥ 13) and weekly hours (≤ 34, 35–45, > 45). Y is income > 50K. Predictors

Table 1. Setting-wise interval for Δ_{nat} under $\mathcal{M}_{\mathrm{cc}}(\Gamma)$, MCAR truth ($\Delta_{\mathrm{nat}} = -0.0253$). "A" means the interval lies below 0 (A better) and "?" means undetermined. The bottom rows give the width ratio of the separate-risk interval $[\min R_A - \max R_B, \max R_A - \min R_B]$ to the direct-difference interval on the same feasible set (P2-FR2-B). Categories: A^s strict, A^w weak (0 attained), $\pm$ both orders; † at least one endpoint is closure-only; A^0 strict for every valid law without a uniform margin. Attainment uses $\max \eta$ with every required $\rho \geq \eta$.

setting	$\Gamma_{cc} = 1$	$\Gamma_{cc} = 3/2$	$\Gamma_{cc} = 2$	no model
A (oracle w)	[-0.025, -0.025] A^s	[-0.033, -0.018] A^s	[-0.038, -0.013] A^s	[-0.051, -0.002] A^s
C (B + masks + unlabelled)	[-0.025, -0.025] A^s	[-0.047, +0.003] $\pm$	[-0.064, +0.023] $\pm$	[-0.111, +0.078] $\pm$
D (conditionals, ρ unknown)	[-0.130, +0.000]† A^0	[-0.272, +0.010]† $\pm$	[-0.355, +0.075]† $\pm$	[-0.518, +0.202]† $\pm$
B (complete cases)	[-0.130, +0.000]† A^w	[-0.272, +0.014]† $\pm$	[-0.355, +0.075]† $\pm$	[-0.518, +0.378]† $\pm$
sep./joint width, A	–	3.52	3.43	3.17
sep./joint width, C	–	2.55	2.45	2.44
sep./joint width, B	2.75	2.11	1.92	1.60

A and B are defined as in Section 7 and use the binned inputs only, so $D(c, r)$ is a fixed loss on the coarse cells. Both were refitted once on the clean training records, with the imputation mode computed on training data only; their tables and hash were stored before evaluation. Feature 1 is missing with probability 0.15. Feature 2 is missing with probability $0.15 + 0.25\,Y + 0.10\mathbb{1}[\text{hours} > 45]$ (MNAR rule) or 0.30 (MCAR control). Instead of drawing masks, we integrate the rule exactly over the evaluation records: $\bar{p}(c, r) = n^{-1} \sum_i \mathbb{1}\{C_i = c\}\pi_r(C_i)$. This law is conditional on these records and is not the super-population. The labels inside π serve only the oracle comparator. The setting-C evaluator receives only complete-case cells with labels and the masses of incomplete observed tuples (schema check in Section G). We report the exact expected-mask analysis conditional on the selected evaluation records. We do not attach a population confidence interval because an appropriate sampling model is not specified for this repaired, development-reused sample. (Hoeffding's inequality itself applies to bounded summands such as $\mathbb{1}\{C_i = c\}\pi_r(C_i)$ whether or not they are integers; the missing ingredient is the sampling model, not integrality.)

Results (Table 2).

- Under the MNAR rule, complete-case dropout ranks B ahead ($+0.0004$), while the risk difference under the rule ranks A ahead (-0.0028). The disagreement survives the split repair. It is small in Brier units, and we claim no practical importance. The MCAR control agrees (-0.0020 for both).
- With the observables of setting C, the MNAR identified sets are empty at $\Gamma_{cc} = 1$ and $5/4$ and admit both orders from $3/2$ on. This agrees with the closed form of Remark 4, $\Gamma_{\min}(o) = 1.311$ for this law, whereas the hidden expected joint has $\Gamma_{\mathrm{true}} = 2.38$. This is no contradiction: other joints with the same observables fit from 1.311 on. An empty set is an exact statement about this conditional law under a too-small Γ_{cc}; it is neither a confidence-set rejection nor evidence that any Γ_{cc} is correct.
- B and D at $\Gamma_{cc} = 1$ behave differently. B cannot refute MCAR and, for the MNAR law, returns "B weakly better", the wrong sign. D, whose information includes the strata's conditional laws, has an empty valid set at $\Gamma_{cc} \leq 5/4$ ($\emptyset^c$): the only LP points give zero mass to a given stratum, so D refutes $\mathcal{M}_{\mathrm{cc}}(1)$ and $\mathcal{M}_{\mathrm{cc}}(5/4)$ like C does. For the MCAR control, B gives A^w and D gives A^0 at $\Gamma_{cc} = 1$. From $3/2$ on (MNAR) or $5/4$ on (MCAR), both orders are admitted.

The development runs on the contaminated split (App. G) showed the same qualitative pattern with different numbers. They used different predictors and a different split, so we do not compare them as estimates of one quantity.

8. Limitations

- **One population.** All numbers come from one binary population with two modalities, two fixed models and the Brier loss. They show that the stated phenomena can occur; they say nothing about how often they occur. The population was seen during development.
- **Real data.** The pilot uses real covariates and labels from one census dataset, but its masks are synthetic, its two "modalities" are grouped tabular features, and there is one repaired record-profile split on already-seen data. Profiles are not persons. The population is the unweighted record population. The effects are small. Results on real multimodal missingness are not reported [TODO: P2-real: natural-missingness dataset, labels only for complete cases, approval and license]. Continuous or high-dimensional features would require discretisation or smoothness assumptions. The resulting bounds would be estimates, not population sharp bounds.
- **Assumptions.** $\mathcal{M}_{\mathrm{cc}}(\Gamma)$ is one of many sensitivity models. Γ is not estimable, and a feasible but misspecified

Table 2. Repaired pilot (P2-PILOT-CLEAN1): expected-mask law on the clean evaluation records. "true Δ" is the risk difference under the mask rule (oracle, hidden labels); "cc dropout" is random dropout on complete cases. Categories of the exact-observable identified set: $\emptyset$ empty; $\emptyset^c$ no valid law (LP closure non-empty); $\pm$ both orders; A^s strict; A^w/B^w weak (tie attained); A^0/B^0 strict for every valid law without a uniform margin.

mask rule	setting	true Δ	cc dropout	$\Gamma_{cc} = 1$	$\Gamma_{cc} = 5/4$	$\Gamma_{cc} = 3/2$	$\Gamma_{cc} = 2$	no model
MNAR	C	-0.0028	+0.0004	$\emptyset$	$\emptyset$	$\pm$	$\pm$	$\pm$
	D			$\emptyset^c$	$\emptyset^c$	$\pm$	$\pm$	$\pm$
	B			B^w	$\pm$	$\pm$	$\pm$	$\pm$
MCAR	C	-0.0020	-0.0020	A^s	$\pm$	$\pm$	$\pm$	$\pm$
	D			A^0	$\pm$	$\pm$	$\pm$	$\pm$
	B			A^w	$\pm$	$\pm$	$\pm$	$\pm$

Γ can certify the wrong sign (Section 5). Assumption 1 excludes labels on incomplete units, and the case of a label indicator independent of the mask is not analysed [TODO: P2-S: label availability not tied to the mask].

- **Inference.** The outer confidence interval is conservative and its width is not analysed. Sharper procedures (Imbens & Manski, 2004; Stoye, 2009) are not implemented [TODO: P2-FR3: valid and less conservative inference; coverage study].

- **Proofs and novelty.** The proofs are elementary and self-checked only. Several ingredients are known (see Section 2), and a forward-citation search of the closest works has not been done [TODO: RW: forward citations of Subbaswamy et al. 2021, Guerdan et al. 2024, Zhou et al. 2023, Choi et al. 2026].

- **Losses.** Only the Brier loss was computed; 0–1 and log loss are not run [TODO: P2-FR1-loss].

9. Conclusion

A missing-modality evaluation identifies little about deployment rankings unless the evaluator also knows how often each modality is missing and what the incomplete units look like. Even then, with labels only on complete cases and a sensitivity parameter above the true one, the natural-missingness Brier difference is identified only when the two models agree on every incomplete input that occurs. Beyond that point, conclusions depend on an unverifiable sensitivity parameter, and in our population they are lost at modest values. In the repaired grouped-feature pilot, complete-case dropout reported a ranking opposite to the risk difference under a label-dependent mask rule. The observable information determined the ranking only in the MCAR control at $\Gamma_{cc} = 1$. Treating complete cases as the population (the oracle setting) can make conclusions look far more robust than the observable information supports. Plug-in feasibility checks and plug-in intervals are not confidence statements. We recommend that missing-modality benchmarks report the mask frequencies and the unlabelled law of incomplete units, and present robustness as a function of an explicit sensitivity parameter rather than as a single dropout score.

Impact Statement

This paper concerns the evaluation of predictive models when inputs are missing for informative reasons, a common situation in clinical and other high-stakes settings. Its main message is cautionary: standard random-dropout evaluations can rank models incorrectly and should not be read as evidence of safety under natural missingness. The analysis uses only synthetic, hand-specified distributions. No personal or patient data were used, and no model was trained. A possible misuse is to present a narrow interval computed under an optimistic sensitivity parameter as a certification. We explicitly show that this can certify the wrong ranking.

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A. Proofs

Lemma 1 (exact linearisation; closure in B/D). Fix r. If $\rho_r > 0$ and (p, λ_r) satisfy (1) multiplied by ρ_r, then $b_r = \rho_r \lambda_r$ satisfies (2) (with $\kappa = v$; with $\kappa = \theta = \rho_{\mathbf{1}} v$ use $b_r = \rho_r \lambda_r / \rho_{\mathbf{1}}$). Conversely, given (p, b_r) with $\rho_r = \sum_c p(c, r) > 0$, set $\lambda_r = b_r / \rho_r$ (resp. $b_r \rho_{\mathbf{1}} / \rho_r$). If $\rho_r = 0$ both descriptions force $p(\cdot, r) = 0$. λ_r enters no other row. In B and D the rows $p(c, \mathbf{1}) = \rho_{\mathbf{1}} v_c$ allow $\rho_{\mathbf{1}} = 0$, under which v is not the complete-case law of any population, and in D the μ-rows allow $\rho_r = 0$ for a stratum whose conditional law is given. The LP set is convex, so if it contains one point z^+ with every required mass positive, every LP point z is the limit of the valid points $(1 - \epsilon)z + \epsilon z^+$, and the LP interval is the closure of the identified interval. If no such z^+ exists (the value of the η program without objective constraint is 0), the valid set is empty and the LP interval is a closure artefact. In B, z^+ always exists ($\rho_{\mathbf{1}} = 1$, no incomplete mass). The endpoint $\min_r \Delta_r^{\mathrm{cc}}$ in Proposition 3(ii) is closure-only when it requires $\rho_{\mathbf{1}} = 0$. □

Theorem 2. If $g \perp (z - z')$ for all $z, z' \in F$, then $g^\top z$ is constant. Conversely, if $g^\top z \neq g^\top z'$ for some z, z', then $g \not\perp z - z'$. For a polyhedron $\{Az = b,\ l \leq z \leq u\}$, $\mathrm{aff}(F) = \{Az = b,\ z_j = \text{const}\ (j \in T)\}$, where T is the set of implicitly tight bounds. So the span of differences is the null space of A augmented with $\{e_j\}_{j \in T}$. Adding e_j for all constant coordinates (a superset of T whose extra rows lie in the row space already) does not change it. □

Proposition 3. At $\Gamma = 1$, (2) gives $p(c, r) = b_r \kappa_c$, so $P(C \mid R = r) \propto \kappa \propto v$. Then $\Delta_{\mathrm{nat}} = \sum_r \rho_r \sum_c v_c D(c, r)$. In C, ρ is fixed by the rows, so Δ_{nat} is identified. Feasibility requires the stated equality of the observed and implied unlabelled laws. In B there is no row on ρ other than the simplex. In D, with $p(c, r) = b_r v_c$ and $\rho_r = b_r$, the μ-rows read $b_r(\sum_{c \in G(r,o)} v_c - \mu_r(o)) = 0$: if the conditional of stratum r matches the complete-case marginal they hold for every b_r and add no restriction; otherwise they force $b_r = \rho_r = 0$, which is not a valid D law. The range over the simplex with the required positivity is the stated set, and its closure is the hull of the Δ_r^{cc}. The attainment claims follow by evaluating at $\rho_{\mathbf{1}} = 1$ (valid in B, not in D) and by convexity of the valid set. (iii) is immediate. □

Proposition 5. Take the lifted polytope in (p, b); in C the rows fix $p(c, \mathbf{1})$, the group sums $\sum_{c \in G(r,o)} p(c, r)$, and hence the total mass. Interiority: for an incomplete mask with $\rho_r > 0$ let $m_r = \min_c e_r(c) > 0$ and $M_r = \max_c e_r(c)$; since $M_r / m_r < \Gamma^2$, any $b_r^* \in (M_r / \Gamma,\ m_r \Gamma)$ gives $b_r^* \theta_c / \Gamma < p^*(c, r) < \Gamma b_r^* \theta_c$ for every c. Hence every direction h with $h(c, \mathbf{1}) = 0$ and $\sum_{c \in G(r,o)} h(c, r) = 0$, together with $\Delta b = 0$, keeps $p^* \pm th$ feasible for small t. Directions that move only b leave p, and hence Δ_{nat}, unchanged. So Δ_{nat} is constant iff $D(\cdot, r)$ is orthogonal to every zero-sum vector on each group, i.e. constant on each group. Under the Brier loss, for $c, c' \in G(r, o)$ with labels 0 and 1, $D(c, r) - D(c', r) = 2\delta(o, r)$, so constancy forces $\delta(o, r) = 0$, and then $u(o, r) = 0$ as well. Conversely, if $\delta = 0$ on all incomplete inputs, then $D(c, r) = 0$ for $r \neq \mathbf{1}$ and $\Delta_{\mathrm{nat}} = \sum_c \theta_c D(c, \mathbf{1})$. □

Remark (oracle A). The additional rows $\sum_r p(c, r) = w_c$ add the functions $a(c)$. Differencing labels at fixed x gives $-2\delta(x_r, r) = a(x, 1) - a(x, 0)$ for every incomplete r. With the all-missing mask present, $\delta(\varnothing) = \alpha(x)$ for every x, so δ is constant on incomplete inputs. The converse holds with $a(x, y) = -2\kappa y$. □

Remark 4. Feasibility: $\rho = e_{\mathbf{1}}$ with $p(c, \mathbf{1}) = v_c$ and $b = 0$ satisfies every row of B and D. Necessity: if a stratum law P_r has $P_r(c)/v_c \in [\lambda/\Gamma, \lambda\Gamma]$ for all c, then its group sums satisfy $r_r(o) = \mu_r(o)/V_r(o) \in [\lambda/\Gamma, \lambda\Gamma]$, so $\max_o r_r(o) / \min_o r_r(o) \leq \Gamma^2$, and $\mu_r(o) > 0$ forces $V_r(o) > 0$. Sufficiency: $P_r(c) = v_c\, r_r(o(c))$ is a law ($\sum_c P_r(c) = \sum_o \mu_r(o) = 1$) with cell ratio $r_r(o(c))$, so with $\lambda_r = \sqrt{\max_o r_r \min_o r_r}$ it lies in $\mathcal{M}_{\mathrm{cc}}(\Gamma)$ exactly when $\Gamma^2 \geq \max_o r_r / \min_o r_r$; then $p(c, r) = \rho_r P_r(c)$ with the observed (C) or any positive (D) ρ matches every row. Hence the infimum is a minimum, the same for C and D, and $\Gamma_{\min}(o) \leq \Gamma_{\mathrm{true}}$ because the true joint is one such law. □

Proposition 7. Every $\mathcal{M}_{\mathrm{cc}}(\Gamma)$ and SUPPORT-ONLY impose $p(c, r) \leq \Gamma b_r \kappa_c = 0$ when $\kappa_c = 0$. □

Proposition 9. Hoeffding's inequality for each cell indicator, followed by a union bound over K cells, gives $P(\vartheta \in B_\alpha) \geq 1 - \alpha$. Outward rounding only enlarges B_α. On this event, any p in the population identified set has $p(c, \mathbf{1}) = \theta_c \in [\kappa_c^{\mathrm{lo}}, \kappa_c^{\mathrm{hi}}]$ and group sums inside the box. With b_r its population scale, $p(c, r) \geq b_r \theta_c / \Gamma \geq b_r \kappa_c^{\mathrm{lo}} / \Gamma$ and $p(c, r) \leq \Gamma b_r \kappa_c^{\mathrm{hi}}$. Summing the lower inequality over c gives $b_r \leq \Gamma / \sum_c \theta_c \leq \Gamma / \sum_c \kappa_c^{\mathrm{lo}}$, so the auxiliary bounds are not active. If $\sum_c \kappa_c^{\mathrm{lo}} = 0$ the bound on b_r is dropped and the incomplete-mask rows reduce to $p(c, r) = 0$ where $\kappa_c^{\mathrm{hi}} = 0$; on the event $\theta_c \leq \kappa_c^{\mathrm{hi}} = 0$ then forces $p(c, r) \leq \Gamma b_r \theta_c = 0$, so p satisfies them. Hence p is feasible for the relaxed program. The event does not involve Γ, so the statement holds for all Γ simultaneously; Δ_{nat} is covered when, in addition, $p^* \in \mathcal{M}_{\mathrm{cc}}(\Gamma)$. □

Status. All proofs above are complete as written but have only been self-checked; an independent statistical reviewer is UNASSIGNED. Open: whether the closure caveat can be removed by a constraint $\rho_{\mathbf{1}} \geq \underline{\rho}$ that is known in practice (UNPROVED); width guarantees for Proposition 9 (UNPROVED); extension to label availability independent of the mask

(NOT STARTED).

B. Linear-Programming Formulation and Checks (P2-FR2)

The program uses the substitution $p(c,r) = b_r\kappa_c/\Gamma + s(c,r)$, $s(c,r) + t(c,r) = (\Gamma\kappa_c^{\mathrm{hi}} - \kappa_c^{\mathrm{lo}}/\Gamma)b_r$, with $s, t \geq 0$ and finite bounds on all auxiliaries (`src/msid/observation.py`). Every endpoint carries a certificate that is checked independently of the pivoting code: either a primal point with exact feasibility and a dual vector whose weak-duality bound equals the objective, or a Farkas vector. The following engineering checks all passed (P2-FR2 manifest):

- every linear program was certified (0 uncertified);
- nesting in information (A $\subseteq$ C $\subseteq$ D $\subseteq$ B) and in Γ;
- whenever the truth satisfies $\mathcal{M}_{\mathrm{cc}}(\Gamma)$, the truth lies in the interval;
- the structural-zero check (Proposition 7);
- the outer interval contains the population interval whenever the box contains the truth;
- the general criterion (Theorem 2) agrees with zero width in all audited cases.

Engineering success is not evidence for the scientific statements.

Table 3. Identification criterion audit on FR1 feasible sets (P2-FR2-G).

set	Γ	width $= 0$	explicit row space	general criterion
pattern	1	True	False	True
pattern	2	False	False	False
observed_law(E1p)	400/211	False	False	False
observed_law(E1p)	2	False	False	False

Table 4. Structural zero: one cell never fully observed (P2-FR2-E).

setting	model	interval	p^* in model	truth in interval
A	2	infeasible	False	None
A	1000	infeasible	False	None
A	SUPPORT ONLY	infeasible	False	None
A	NO MODEL	[-0.094, -0.046]	True	True
C	2	infeasible	False	None
C	1000	[-0.154, +0.135]	False	True
C	SUPPORT ONLY	[-0.154, +0.135]	False	True
C	NO MODEL	[-0.154, +0.135]	True	True

Secondary environments (P2-FR2-C): whenever $\Gamma \geq \Gamma_{cc}$ the interval contains the truth. Below Γ_{cc}, sets are often empty, but not always. For E1m, E2p and E2m in setting C at $\Gamma = 3/2$ or 2, the set is non-empty and the interval excludes the truth. For E2m at $\Gamma = 3/2$ it certifies the wrong sign.

C. Earlier Oracle Analysis (P2-FR1)

FR1 assumed the full-data law w known and used the policy box $q_r/\Gamma_{\mathrm{box}} \leq \pi(r \mid c) \leq \min(1, \Gamma_{\mathrm{box}} q_r)$ around the dropout law (reference: q, i.e. with matched pattern rates $P(c \mid R = r)/w_c \in [\Gamma_{\mathrm{box}}^{-1}, \Gamma_{\mathrm{box}}]$ with scale fixed to 1), with optional feature-rate, pattern-rate or unlabelled-law constraints (labels observed nowhere). Its rows labelled "$\Gamma = \infty$" are the *no-box* set. They coincide with the closure of the union over finite Γ when $q > 0$, but in the support-violation check (where q gives mask 00 zero probability) they amount to deleting the support restriction, not to taking $\Gamma \to \infty$.

FR1 statements, with the corrections made in this version:

Table 5. FR1: joint versus separate interval and a hand-specified scenario grid, pattern-rate constraint (P2-FR1-B). "A" means A better and "?" means undetermined.

Γ	joint interval	separate interval	scenario grid in set
5/4	[-0.0430, -0.0092] (A)	[-0.0559, +0.0028] (?)	[-0.0253, -0.0253]
3/2	[-0.0552, +0.0026] (?)	[-0.0817, +0.0242] (?)	[-0.0253, -0.0253]
2	[-0.0707, +0.0192] (?)	[-0.1214, +0.0568] (?)	[-0.0580, -0.0253]
3	[-0.0871, +0.0406] (?)	[-0.1708, +0.1013] (?)	[-0.0580, +0.0075]

Table 6. FR1: ranking-transfer threshold Γ^*_{box} (smallest Γ_{box} whose interval contains 0) by information set, and the closed-form lower bound (P2-FR1-E).

information (FR1 model, w known)	Γ^* bracket
no constraint	(1.3966, 1.3967]
feature rates	(1.4243, 1.4243]
pattern rates	(1.4390, 1.4391]
unlabelled law	(2.0692, 2.0693]
closed form Γ_{cf} (pattern)	1.2689

- **Attainability and sharpness relative to the set.** The endpoints are attained by valid policies, and the interval is sharp relative to the stated assumptions.
- **Row-space characterisation.** It needs the relative-interior assumption. The $\Gamma = 1$ row of Table 3 shows the explicit test failing at the boundary.
- **Brier corollary.** Here δ constant is required on all inputs, because labels are observed nowhere. It needs both labels at every x, the all-missing mask with positive probability, and interiority.
- **Closed-form bound.** Under matched pattern rates, $|\Delta(\pi) - \Delta(q)| \leq (\Gamma - 1)\,\mathbb{E}_{w\otimes q}|D_{\mathrm{int}}|$, where D_{int} is the cell-by-mask interaction component.
- **Unlabelled data and Γ.** They can refute small Γ but never bound it from above.

FR1 checks, all PASS: certificates; $\Gamma = 1$ point; nesting; membership implies containment; support violation reported as infeasible (not a number); rank test against width, 36/36; hand values of Example 8. In FR1 the plug-in containment frequencies were 8/100 and 5/100. These are not coverage.

D. Additional Toy Results

E. Comparison with the Two Closest Works

Table 10 compares the two closest prior works with this paper along the five aspects that matter for our question. Entries for this paper are tagged [KNOWN] (ingredient taken from prior work), [ADAPTED] (prior idea in a new observation model) or [UNVERIFIED novelty] (no forward-citation search has been done). The prior-work entries were checked against the original texts as recorded in the repository's related-work notes (main bodies and the cited appendix parts read from the arXiv/ar5iv text, key sentences grep-checked against the PMLR PDFs); equation and definition numbers refer to the arXiv versions.

F. Admission Conditions for Natural-Missingness Data

A natural-missingness follow-up needs:

- the original missingness record, not an imputed one;
- outcomes for incomplete units that are independent of the evaluator's input, so that Δ_{nat} can be checked;
- the complete-case raw variables, with common support between complete and incomplete units;

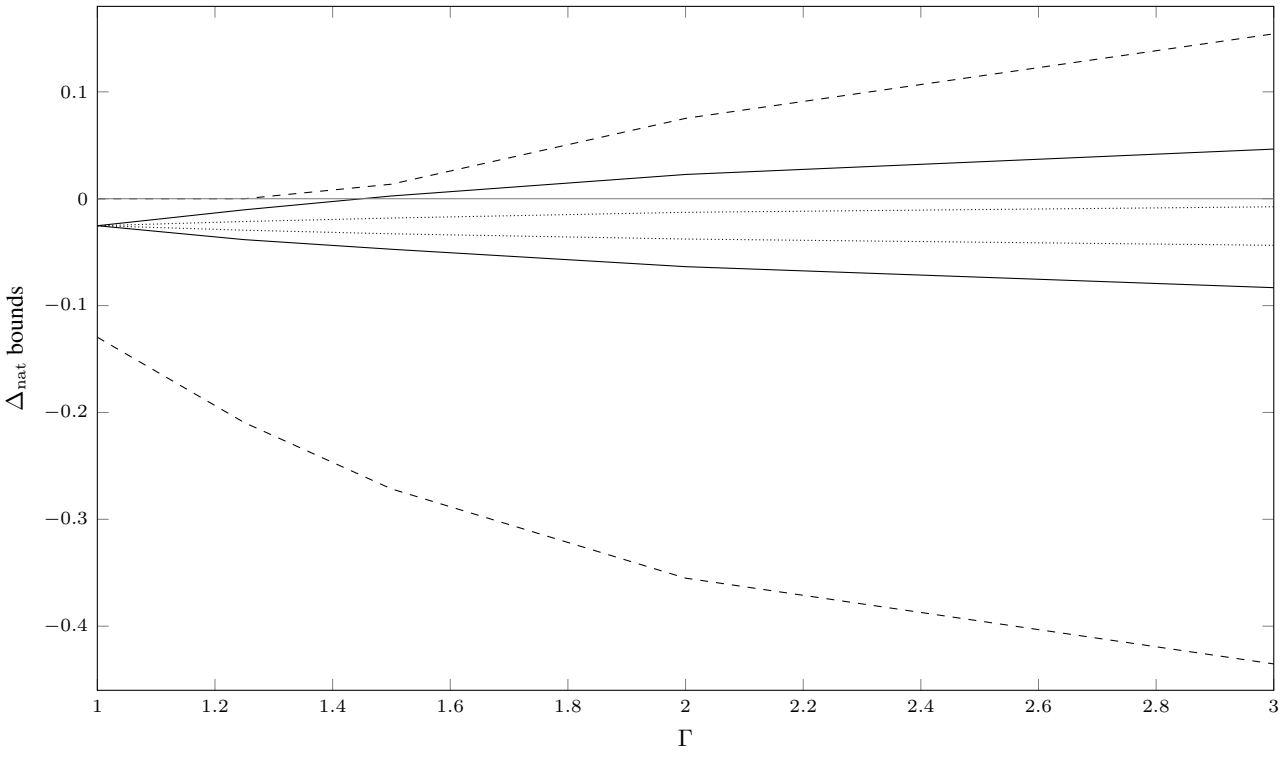


Figure 1. Lower and upper bounds of Δ_{nat} versus Γ for settings A (dotted), C (solid) and B (dashed); MCAR truth (P2-FR2-B).

- evidence of the entity unit, i.e. person or visit identifiers for isolation;
- a licence or permission that allows local analysis;
- a binned estimand for which $D(c, r)$ is fixed, or an explicit outer bound.

No such dataset has been selected or downloaded.

G. Development Log and Corrections

- **v1.** B/D cells at $\Gamma_{cc} \in \{1, 5/4\}$ were called "undetermined".
- **v2.** These cells were re-labelled "weak" using a check of $\rho_{\mathbf{1}} > 0$ only.
- **v3.** Checking every definition-required stratum ($\max \eta$ LP, P2-PILOT2) keeps "weak" for B and changes D to "strict without margin".
- $\mathcal{M}_{\mathrm{cc}}$ **definition.** v1 omitted absolute continuity, which the programs always imposed.
- **Parameters.** Γ_{box} (FR1) and Γ_{cc} (FR2) use different references and are not compared.
- **Data unit.** v2 called attribute-tuple groups "entities"; the v3 provenance check shows they are not verified persons.
- **v4.2: order of checks.** v3 computed relaxed bounds first and checked endpoints afterwards, which can report an interval for an empty set of valid laws (the CLEAN1 D rows at $\Gamma_{cc} \leq 5/4$). The analysis now runs valid-set check, then bounds, then endpoints, with per-setting required strata and exact rational certificates; the related-work table and the development runs were not changed.
- **v4: split repair.** The PILOT1/2 split key contained the raw income string, and 956 covariate profiles crossed the split. Those runs are kept as DEVELOPMENT_WITH_PROFILE_OVERLAP (Table 11). They are valid algebraic finite-population examples for their fixed predictors, but they are not clean held-out evidence. P2-PILOT-CLEAN1

Table 7. Oracle values (target labels, true policy; scoring only) versus the two dropout estimands, and the true $\mathcal{M}_{\mathrm{cc}}$ parameter (P2-FR2-A). Z_check is the constructed support violation.

environment	Γ_{cc}	Δ_{nat} (oracle)	dropout on complete cases	dropout on population
E0_MCAR	1.00	-0.0253	-0.0253	-0.0253
E1p_self_x2	2.19	-0.0461	-0.0156	-0.0253
E1m_self_x2	2.33	-0.0044	-0.0349	-0.0253
E2p_label_x2	2.20	-0.0580	-0.0070	-0.0253
E2m_label_x2	2.32	+0.0075	-0.0436	-0.0253
E3p_self_x2_strong	4.59	-0.0600	-0.0091	-0.0253
E3m_self_x2_strong	10.08	+0.0095	-0.0414	-0.0253
E4_correlated_mcar	1.00	-0.0300	-0.0253	-0.0253
Z_check	∞	-0.0689	+0.0088	-0.0253

Table 8. Finite samples in setting C, MCAR truth; 2 draws per n, shown as illustrations only (P2-FR2-F). "infeas." means the plug-in program is empty.

n	Γ	plug-in (draw 1 / draw 2)	outer 95% CI (draw 1)	population
1000	1	infeas. / infeas.	[-0.130, +0.052] ?	[-0.025, -0.025] A
1000	2	? / ?	[-0.161, +0.102] ?	[-0.064, +0.023] ?
10000	1	infeas. / infeas.	[-0.058, +0.003] ?	[-0.025, -0.025] A
10000	2	? / ?	[-0.101, +0.055] ?	[-0.064, +0.023] ?
100000	1	infeas. / infeas.	[-0.035, -0.016] A	[-0.025, -0.025] A
100000	2	? / ?	[-0.076, +0.032] ?	[-0.064, +0.023] ?

repairs label parsing and group allocation and refits the predictors once. It found 0 unresolved labels and 0 profile leakage, and the evaluator schema check still holds.

All raw outputs are unchanged. Corrections are derived files (`results/derived/`) or new raw files of separately labelled analyses.

H. Reproducibility

Code: `src/msid`. Drivers: `scripts/run_p2_first_run.py` and `scripts/run_p2_fr2.py`. Configs: `configs/`. Raw outputs and manifests: `results/raw/`. Tables: `scripts/make_paper_tables.py`. Only the Python standard library is used. Claims are mapped to evidence in `paper/claims.csv`.

Table 9. Pilot, setting C: finite-population identified sets and outer 95% intervals by Γ_{cc} ($\emptyset$ empty; $\pm$ both orders; – not in the pre-registered outer grid).

mask rule	set	$\Gamma_{cc} = 1$	$\Gamma_{cc} = 5/4$	$\Gamma_{cc} = 3/2$	$\Gamma_{cc} = 2$	$\Gamma_{cc} = 3$	no model
mnar	identified	$\emptyset$	$\emptyset$	[-0.005, +0.004] $\pm$	[-0.009, +0.006] $\pm$	[-0.016, +0.008] $\pm$	[-0.027, +0.009] $\pm$
mnar	outer 95%	[-0.007, +0.007] $\pm$	–	[-0.017, +0.010] $\pm$	[-0.022, +0.011] $\pm$	–	[-0.036, +0.012] $\pm$
mcar	identified	$\emptyset$	[-0.005, +0.001] $\pm$	[-0.008, +0.003] $\pm$	[-0.013, +0.005] $\pm$	[-0.019, +0.007] $\pm$	[-0.028, +0.009] $\pm$
mcar	outer 95%	[-0.011, +0.005] $\pm$	–	[-0.021, +0.009] $\pm$	[-0.026, +0.010] $\pm$	–	[-0.037, +0.012] $\pm$

Table 10. Observation, estimand, sensitivity model, sharpness and sampling uncertainty in Subbaswamy et al. (2021), Guerdan et al. (2024) and this paper.

Aspect	Subbaswamy et al. (2021)	Guerdan et al. (2024)	This paper
Observed information	One i.i.d. test sample with X, Y fully observed; analyst splits variables into immutable Z, mutable W, dependent V; the shift replaces $P(W \mid Z)$ by $Q(W \mid Z)$ with $P(Z)$ and $P(V \mid W, Z)$ fixed (Sec. 2.1).	Selective labels: Y observed only when the status-quo decision $D^{\pi_0} = 1$; data $(X, T^{\pi}, D^{\pi_0}, Y)$ from the status-quo policy; proposed policy π known (Sec. 3).	[ADAPTED] Complete cases with labels (θ_c) plus unlabelled incomplete tuples $\omega(r, o) = P(R = r, X_r = o)$ (setting C); less in B/D; full-data law only in oracle A. Their (W, Z)-shift with $W = R$, $Z = (X, Y)$, but conditioned on what is observable under masking.
Comparison estimand	Worst-case risk $R_{\alpha,0}$ of *one* fixed model: expected loss on the worst $(1 - \alpha)$-subsample (Eq. 1–2); two models are compared by their separate worst-case curves (Fig. 4a).	Regret $m^*(\pi) - m^*(\pi_0)$; Def. 4.1 baseline interval subtracts separate bounds, Def. 4.2 δ-regret optimises the difference jointly over $\mathcal{V}(p)$; Thm. 4.3 quantifies the gain.	[KNOWN] Risk difference $\Delta_{\text{nat}} = \sum_{c,r} p(c, r) D(c, r)$ of two fixed predictors under the same unknown natural-mask law, optimised jointly over p by two LPs; the cancellation is an instance of their Def. 4.2 and is not claimed as new.
Sensitivity / uncertainty model	One-sided likelihood-ratio ball on $P(W \mid Z)$: $dQ/dP \leq 1/(1 - \alpha)$ a.e. (App. Eq. 11–12; D_∞ ball, Remark 3); $Q \ll P$ by construction; α user-chosen.	Pointwise bounding functions $\underline{\tau}(x) \leq \mathbb{E}[Y(1) \mid D = 0, X = x] \leq \overline{\tau}(x)$ (Asm. 5.1), implied by MSM/Rosenbaum/IV/proximal; e.g. MSM gives $\Lambda^{-1}\mu_1 \leq \mu_0 \leq \Lambda\mu_1$ (Lemma B.4); yields the box $\mathcal{V}(p; \tau)$.	[ADAPTED] Pattern-mixture bounded ratio $\lambda_r/\Gamma \leq P(C = c \mid R = r)/v_c \leq \lambda_r\Gamma$ relative to the complete-case law, free scale λ_r per mask, absolute continuity ($\mathcal{M}_{cc}(\Gamma)$, two-sided); Γ analyst-chosen, not estimable.
Sharpness claim	$R_{\alpha,0}$ is point-identified; the sup in Eq. 1 equals the conditional-quantile (CVaR-type) dual, Eq. 3–4; no partial-identification sharpness question arises.	Lemma 5.2 bounds are sharp given τ (App. B.6.1); Thm. 5.3: $\mathcal{V}(p; \tau)$ is minimal given τ; App. B.1/B.6.2 state this does *not* guarantee the tightest regret interval under the causal assumption itself.	[UNVERIFIED novelty] LP interval is the closure of the identified set of Δ_{nat} under the stated assumptions when a valid law exists; attained vs closure-only endpoints separated by one extra LP; exact observable threshold $\Gamma_{\min}(o)$ in closed form (C, D) with $\Gamma_{\min}(o) \leq \Gamma_{\text{true}}$.
Sampling uncertainty	Cross-fitted debiased estimator (Alg. 1, Eq. 5); Thm. 1: $\sqrt{N}$-consistent and asymptotically normal; Wald CIs from Eq. 7; variance scales as $1/(1 - \alpha)^2$; discrete W needs noise augmentation (Remark 6).	Cross-fitted plug-in (Alg. 1; Thm. 6.1: error inherits nuisance rates $+O_P(n^{-1/2})$); doubly robust variant for the MSM (Eq. 2, App. C); 95% bootstrap CIs around interval endpoints in experiments.	[ADAPTED] Conservative outer interval: Hoeffding/Bonferroni box on the observable multinomial, relaxed LP; on one event it covers the identified interval for all Γ simultaneously and covers Δ_{nat} for $\Gamma \geq \Gamma_{\text{true}}$; it does not validate Γ; not sharp, no width analysis.
Not claimed here	Outside their scope: a bound on the *difference* between two models under a common shift; a two-sided ratio; partially observed features.	Outside their scope: feature masks; constraints from unlabelled incomplete observations; attained-vs-closure endpoint distinction.	[KNOWN] LP bounds, the direct-difference bound, joint-versus-separate cancellation, bounded-ratio sensitivity models and tipping-point Γ^* summaries are prior art; no $\sqrt{n}$ or doubly robust inference, no data-based validation of Γ, no forward-citation novelty check.

Table 11. DEVELOPMENT_WITH_PROFILE_OVERLAP (P2-PILOT1/2, superseded split). Pilot comparison for the same predictors, data and Γ_{cc} grid. *expected*: noise-free mask law; *realised*: the one pre-registered mask draw (finite-population identified sets); *outer*: conservative 95% interval from the realised sample. The true Δ_{nat} uses hidden labels. $\emptyset$ empty; $\pm$ both orders; A^s strict; – not in the pre-registered outer grid. Γ_{cc} column: smallest free-scale value containing the law (P2-PILOT1, P2-PILOT2).

mask rule	law	true Δ_{nat}	cc dropout	Γ_{cc}	$\Gamma_{cc} = 1$	$\Gamma_{cc} = 5/4$	$\Gamma_{cc} = 3/2$	$\Gamma_{cc} = 2$	no model
MNAR	expected	-0.0033	+0.0006	2.38	$\emptyset$	$\emptyset$	$\pm$	$\pm$	$\pm$
	realised	-0.0038	+0.0006	2.85	$\emptyset$	$\emptyset$	$\pm$	$\pm$	$\pm$
	outer 95%				$\pm$	–	$\pm$	$\pm$	$\pm$
MCAR	expected	-0.0020	-0.0020	1.00	A^s	$\pm$	$\pm$	$\pm$	$\pm$
	realised	-0.0025	-0.0020	1.79	$\emptyset$	$\pm$	$\pm$	$\pm$	$\pm$
	outer 95%				$\pm$	–	$\pm$	$\pm$	$\pm$